
LATENT PLANNING VIA EMBEDDING ARITHMETIC: A CONTRASTIVE APPROACH TO STRATEGIC REASONING

Anonymous authors
Anonymous submission

ABSTRACT

Planning in high-dimensional decision spaces can be made substantially more direct by moving the planning problem into learned representations. Rather than training policies or value heads, we show that planning can be performed inside an evaluation-aligned embedding space. We introduce SOLIS, a contrastively trained system that learns this space explicitly: outcome similarity is encoded by proximity, and a single global *advantage vector* organizes the representation from losing to winning regions. Candidate actions are ranked by their alignment with this direction, reducing action selection to decisive vector operations in latent space. In chess, SOLIS reaches competitive strength under constrained conditions using only shallow embedding-guided search. These results establish evaluation-aligned latent planning as a lightweight and effective alternative to traditional dynamics models or policy learning in settings where reliable evaluations are available.

1 INTRODUCTION

Planning in high-dimensional state spaces remains one of the central bottlenecks in artificial intelligence. Traditional reinforcement learning systems depend on explicit search or learned policies, and both approaches can become inefficient as the action space expands (Jiang et al., 2023; Gieselmann and Pokorný, 2022). Latent planning directly addresses this bottleneck by reasoning over compact learned representations rather than raw states, reducing complexity while preserving the structure needed for strategic decision-making (Hafner et al., 2019; 2020; Schrittwieser et al., 2020).

We take the next step and make the embedding space itself the planning substrate. **SOLIS** (*Strategic Optimization via Latent Interpolative Spaces*) is a chess engine built around a transformer encoder trained with supervised contrastive learning to embed states into an evaluation-aligned latent space. In this space, distance captures outcome similarity, and directional movement provides a practical signal for progress toward favorable states. SOLIS exploits this geometry by ranking candidate moves according to their alignment with a single global “advantage vector” that orients the latent space from losing to winning regions.

SOLIS delivers strong results even at modest search depths. With only a 5-ply search under 50 ms/move blitz conditions, it achieves an Elo of 2500+; in equal-depth matches at search depth $S \in \{3, 4, 5\}$ with a fixed branching factor, it competes directly with the configured Stockfish baseline in our setup. Visualizations of real games further show smooth trajectories through the embedding space, making the latent planning process qualitatively interpretable.

This contribution is deliberately distinct from latent planning methods that couple learned dynamics with policy or value heads (Hafner et al., 2019; 2020; Schrittwieser et al., 2020): SOLIS performs action selection using only the geometry of an evaluation-aligned representation. Chess serves as a stringent testbed, and the same vector-based planning principle naturally applies to other perfect-information domains where state evaluations are available or learnable.

2 BACKGROUND

Planning in very large action spaces is difficult because the number of possible action sequences grows exponentially with each additional decision step. This combinatorial pressure makes the choice of representation decisive: a useful representation must compress the space while retaining

the structure needed for action selection. This section establishes the decision-process, chess-engine, and contrastive-learning foundations that motivate SOLIS.

2.1 MARKOV DECISION PROCESSES

Sequential decision-making tasks are commonly formalized as Markov Decision Processes (MDPs), tuples $\{\mathcal{S}, \mathcal{A}, T, R, p(s_0), \gamma\}$ in which $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, T is the transition model, R is the reward function, $p(s_0)$ is the initial state distribution, and $\gamma \in [0, 1]$ discounts future rewards (Puterman, 2014). A policy $\pi : \mathcal{S} \rightarrow p(\mathcal{A})$ maps states to probability distributions over actions.

Within this framework, a *model* provides reversible access to the MDP dynamics: given a state–action pair, it returns the next state and reward (Moerland et al., 2022). Games such as chess and Go provide this model exactly through their rules. When rules are unavailable, the dynamics must instead be approximated from data (Gieselmann and Pokorny, 2022).

Planning and reinforcement learning differ in how they exploit access to MDP dynamics (Moerland et al., 2022). Planning methods use a model to simulate continuations from the current state and select an action, mirroring the deliberate mental simulation used in human planning (Russell and Norvig, 1995). The resulting solution is *local*: it is tailored to the current state and discarded after execution. Reinforcement learning methods, by contrast, generally lack reversible access to the dynamics (Sutton and Barto, 1998). They interact with the environment step by step and therefore learn a *global solution*, such as a policy or value function, that generalizes across the state space.

Model-based reinforcement learning combines these regimes by using a model while also learning a global solution. Following Moerland et al. (2022), this setting has two principal forms. In model-based RL with a learned model, as in Dyna (Sutton, 1991), both the dynamics and the global solution are learned. In model-based RL with a known model, as in AlphaZero (Silver et al., 2017), the dynamics are given and learning focuses on the global policy or value representation.

2.1.1 MODEL-BASED RL WITH A LEARNED MODEL

Much of model-based RL belongs to the learned-model setting, where the agent learns both dynamics and a global decision rule. PlaNet (Hafner et al., 2019) learns a latent dynamics model and couples it to a learned global solution. Dreamer (Hafner et al., 2020) and its successors DreamerV2 (Hafner et al., 2022) and DreamerV3 (Hafner et al., 2024) scale this recipe to more complex tasks while retaining global value functions and policies. MuZero (Schrittwieser et al., 2020) also learns latent dynamics, but strengthens action selection by integrating them with tree search while training a global policy and value function. Dyna (Sutton, 1991) and MBPO (Janner et al., 2021) likewise use learned models to support value or policy learning.

2.1.2 MODEL-BASED RL WITH A KNOWN MODEL

When environment dynamics are available, the algorithm can concentrate on coupling exact search with learned global guidance. AlphaZero (Silver et al., 2017) uses the game rules as its model and combines Monte Carlo tree search (MCTS) with learned policy and value heads. SAVE (Hamrick et al., 2020) similarly assumes a simulator and integrates Q-learning with MCTS: a learned Q-function guides search, and search-produced Q-estimates are amortized back into the network.

2.1.3 LATENT PLANNING

Latent planning makes decision-making tractable by operating in compressed representation spaces rather than raw state spaces. PlaNet (Hafner et al., 2019), Dreamer (Hafner et al., 2020), and MuZero (Schrittwieser et al., 2020) pair latent dynamics models with value or policy heads so that simulations can proceed in learned space. Expansive Latent Space Trees (ELAST) (Gieselmann and Pokorny, 2022) formulate latent planning as exploratory tree search, combining contrastive representations with sampling-based expansion for long-horizon control. Trajectory Autoencoding Planner (TAP) (Jiang et al., 2023) similarly learns a compact latent action space and plans over discrete action codes.

SOLIS sharply separates itself from this pattern. It trains no policy head, no value head, and no dynamics model; instead, it distills evaluation targets directly into the geometry of the representation. At inference, planning proceeds by moving through regions aligned with favorable outcomes, using the representation itself as the decision surface.

2.2 CHESS ENGINES

IBM demonstrated the power of scaled chess computation in 1997, when Deep Blue defeated reigning world champion Garry Kasparov in a head-to-head match. Early versions of Stockfish (Romstad et al., 2008) followed the same broad paradigm: handcrafted evaluation functions paired with deep alpha-beta tree search. This paradigm achieved superhuman performance, but its ceiling was tied to the chess expertise encoded by the designers of the static evaluation function.

Those bottlenecks motivated engines that learn evaluation rather than inheriting it from human-authored heuristics. AlphaZero (Silver et al., 2017) and its open-source counterpart Leela Chess Zero (Pascutto and Linscott, 2018) decisively shifted the field toward systems that learn position evaluation through self-play reinforcement learning with no prior knowledge beyond the rules of chess. Modern Stockfish versions likewise adopted an efficiently updatable neural network evaluation module (NNUE) (Nasu, 2018), improving evaluation while preserving traditional alpha-beta search.

These advances produced engines far beyond human strength, yet the underlying search machinery remains recognizably rooted in principles introduced by v. Neumann (1928) and Shannon (1950) and later instantiated in Deep Blue. Strong engines still rely on deep alpha-beta search through vast numbers of states (Lai, 2015). SOLIS targets this search bottleneck directly by using contrastive learning to make planning a traversal problem in a high-dimensional evaluation-aligned embedding space.

2.3 CONTRASTIVE LEARNING

Contrastive representation learning gives SOLIS the mechanism for imposing evaluation geometry. It trains a model to place similar inputs close together and dissimilar inputs farther apart in latent space (Chen et al., 2020). Given a batch I of training examples containing an anchor input x_i , an encoder $f(\cdot)$ maps the anchor to a normalized embedding $z_i = f(x_i)$. Positive pairs (z_i, z_j) denote semantically similar inputs, while the remaining batch samples serve as negatives.

A standard contrastive objective is InfoNCE (van den Oord et al., 2019), which makes the attraction to positives explicit:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k \in A(i)} \exp(\text{sim}(z_i, z_k)/\tau)}, \quad (1)$$

Here $A(i) = I \setminus \{i\}$ is the set of all batch samples except anchor i , $\text{sim}(\cdot, \cdot)$ is the similarity metric, and τ is a scalar temperature parameter. In self-supervised contrastive learning, the positive sample z_j is an augmentation of anchor z_i , so labels are unnecessary (Chen et al., 2020; Khosla et al., 2021).

When labels are available, supervised contrastive loss (SupCon) (Khosla et al., 2021) strengthens InfoNCE by allowing multiple positives for each anchor. The SupCon objective is

$$\mathcal{L}_{\text{sup}} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a \in A(i)} \exp(\text{sim}(z_i, z_a)/\tau)}, \quad (2)$$

where $P(i)$ is the set of positives for anchor i , and $A(i)$ is defined as above. Increasing the number of positive and negative pairs improves representation quality (Khosla et al., 2021; Chowdhury et al., 2024), making SupCon a natural fit for constructing an evaluation-ordered chess state space.

3 METHODS

We now specify the SOLIS pipeline: tokenization, transformer encoding, supervised contrastive training, and embedding-guided action selection at inference.

3.1 INPUT REPRESENTATION AND TOKENIZATION

Each chess position is represented by its Forsyth-Edwards Notation (FEN) ASCII string, which compactly encodes piece placement, side to move, castling rights, en passant target, half-move clock, and full-move counter. Following Ruoss et al. (2024), we expand run-length encodings and tokenize each FEN into a fixed-length sequence of 77 tokens. This representation is intentionally compact, but it also makes the problem non-Markovian because historical information relevant to rare rules such as threefold repetition or the fifty-move rule is omitted (Ruoss et al., 2024). Section 5 addresses the resulting limitation directly.

3.2 ENCODER ARCHITECTURE

The SOLIS encoder is a multi-layer transformer (Vaswani et al., 2023) adapted to tokenized chess positions. We train two deliberately sized variants, SOLIS Base and SOLIS Mini, to test whether evaluation-aligned planning survives model compression. Table 1 summarizes the exact configurations used in our experiments.

Table 1: Transformer configurations used to evaluate SOLIS across Base and Mini capacity regimes.

Model	Layers	Embedding dim. D	Heads	MLP Size	Params
SOLIS Mini	6	128	8	256	0.8M
SOLIS Base	6	1024	16	1024	41M

Each input sequence is mapped to D -dimensional vectors using a learned token embedding matrix. A classification token (CLS) (Devlin et al., 2019; Dosovitskiy et al., 2021) is prepended so the network has a dedicated summary representation for the full board state. Because FEN token sequences have fixed length, we add learned positional encodings to the token embeddings (Ruoss et al., 2024). Stacked transformer encoder layers with GELU (Hendrycks and Gimpel, 2023) activations and dropout (Srivastava et al., 2014) process the sequence, after which the final CLS hidden state is extracted, linearly projected, and ℓ_2 normalized to produce the planning embedding.

3.3 SUPERVISED CONTRASTIVE TRAINING

We train the encoder on 5 million randomly sampled positions from ChessBench (Ruoss et al., 2024), each annotated with a Stockfish-evaluated win probability for the player to move. We normalize every evaluation to White’s perspective, so values near 1.0 denote a decisive White advantage and values near 0.0 denote a decisive Black advantage.

Positive samples are defined by evaluation agreement. We set δ as the margin for treating two positions as similar, precompute all positions whose win probabilities differ from an anchor by less than δ , and randomly sample five positives during training. For the batch-level mask, pairs are marked positive whenever their evaluation difference is below δ . This margin-based construction pulls together states with nearly equal evaluations, pushes apart positions with larger evaluation gaps, and explicitly induces an ordering from losing to winning regions.

We optimize this geometry with supervised contrastive learning (SupCon) (Khosla et al., 2021) using the constructed batch masks to define positives and negatives. Pairwise scores use cosine similarity, and temperature $\tau = 0.07$ (Girdhar et al., 2023; Radford et al., 2021) controls distribution sharpness. Optimization uses stochastic gradient descent with momentum 0.9 (Dosovitskiy et al., 2021; Chen et al., 2020; He et al., 2015), batch size 128, for 400,000 steps. Training runs on six NVIDIA L40S GPUs (48 GB each) with data parallelism (Paszke et al., 2019). Appendix A.2 provides the complete hyperparameter specification.

Figure 1 visualizes the learned SOLIS Base representation with a 2D UMAP projection (McInnes et al., 2020). Each point is an encoded test position colored by Stockfish-evaluated win probability for White. Strong White advantages appear in gold (*White has checkmate*), while strong Black advantages appear in dark green (*Black has checkmate*), confirming that the learned space organizes positions by evaluation rather than merely by surface board features.

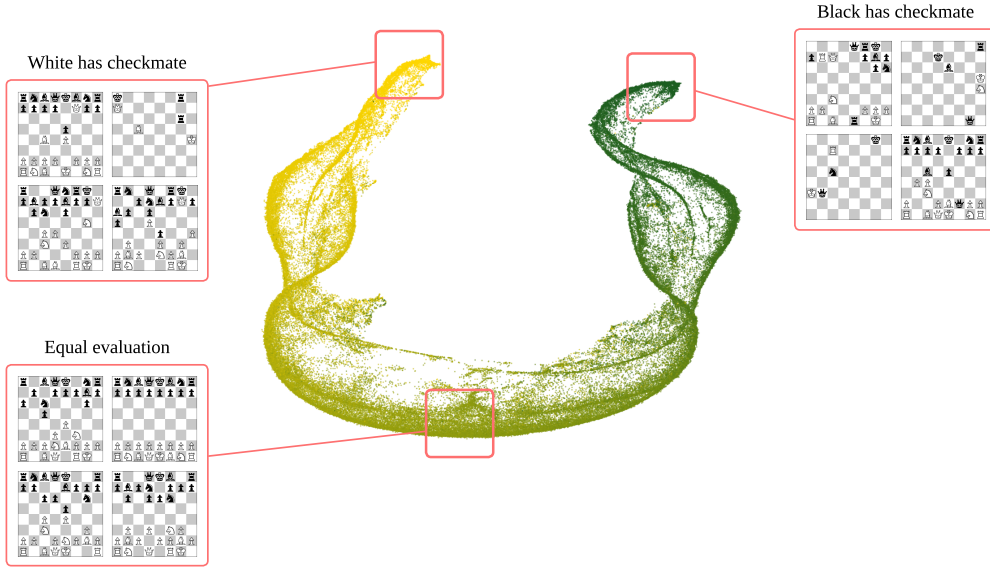


Figure 1: UMAP projection of the SOLIS Base embedding space, showing a clear evaluation-aligned organization from Black-favored regions (green) to White-favored regions (gold).

3.4 EMBEDDING-GUIDED SEARCH

At inference, SOLIS turns this learned geometry into a planner by ranking legal continuations along a single global advantage direction. We compute mean embeddings from held-out positions at the extremes of evaluation,

$$\mu_{\text{White}} = \mathbb{E}[z \mid p=1.0], \quad \mu_{\text{Black}} = \mathbb{E}[z \mid p=0.0],$$

and define the normalized advantage vector

$$\vec{a} = \frac{\mu_{\text{White}} - \mu_{\text{Black}}}{\|\mu_{\text{White}} - \mu_{\text{Black}}\|}.$$

We evaluate two direct mechanisms for converting this vector into scalar move scores.

3.4.1 UNANCHORED (GLOBAL-DIRECTIONAL)

The unanchored method scores each child embedding by direct alignment with the advantage direction. For child embedding z' , the score is

$$\text{score}_{\text{dir}}(z') = \langle z', \vec{a} \rangle,$$

This favors moves whose embeddings lie furthest along the global advantage direction, independent of the origin. Figure 2 illustrates this mechanism.

3.4.2 ANCHORED (RELATIVE PROJECTION)

The anchored method removes origin dependence by shifting each child embedding relative to the Black mean before scoring:

$$\text{score}_{\text{anch}}(z') = \langle z' - \mu_{\text{Black}}, \vec{a} \rangle,$$

The resulting value is proportional to signed distance from μ_{Black} along $\vec{a}$. This anchoring eliminates reliance on an arbitrary origin and, in practice, yields a more stable basis for comparing continuations.

Both scoring mechanisms use the same search procedure (Figure 2). From the “Starting board state”, SOLIS “Extract[s] possible moves” and embeds each child in parallel with the “Transformer

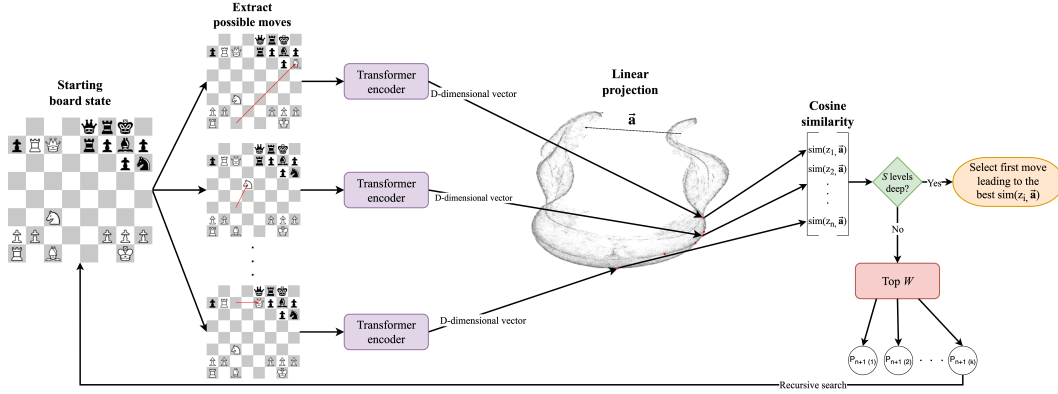


Figure 2: System overview of SOLIS. Candidate moves are embedded, scored by alignment with the global advantage vector, and pruned to a compact set of high-value continuations. For clarity, the diagram shows the unanchored scoring mechanism $\cos(z', \vec{a})$.

encoder”. It then computes either $\text{score}_{\text{dir}}$ or $\text{score}_{\text{anch}}$ in the “Cosine similarity” block and keeps the “Top- W ” children. Once the search reaches “ S levels deep?”, SOLIS terminates and “Select[s] first move. . .”; otherwise it continues “Recursive search” with min-max preference, where White maximizes and Black minimizes. Zobrist hashing (Zobrist, 1990) supplies a transposition table that prevents repeated evaluation of identical positions.

4 RESULTS

4.1 RATING CALCULATION

We estimate Elo by evaluating each model against Stockfish under a fixed 50 ms per-move time limit (Ruoss et al., 2024). Appendix A.3 gives the full Stockfish configuration. The evaluation ablates both search breadth and search depth: for each depth/width setting, SOLIS plays 600+ games with alternating colors against Stockfish at multiple Elo caps. Both unanchored and anchored planners are evaluated in this protocol. Caps are chosen so that each model configuration includes at least one matchup with a positive win rate and one with a negative win rate, making the rating estimate identifiable. Elo ratings and 95% CIs are estimated with Bayesian logistic regression using BayesElo (Coulom, 2008) and its default confidence parameter of 0.5. Figure 3 reports unanchored ratings, Figure 4 reports anchored ratings, and Appendix A.6 provides the full match tables for independent Elo verification.

Unanchored ratings Unanchored planning scales reliably with search. Depth delivers the largest gains, while width produces a clear improvement from $W \in \{1, 2\}$ to $W=3$ and then largely saturates at $W=5$. This saturation shows that the learned embedding already ranks the strongest continuations near the top, so broader expansion contributes little additional strength. At shallow depths, SOLIS Base leads Mini by roughly 100–250 Elo; at depths 4 and 5, Mini remains competitive with Base and even surpasses it at $W \in \{3, 5\}$ under this scoring rule.

Anchored ratings Anchored scoring produces the cleanest scaling pattern: Elo rises smoothly with depth and width, and the Base–Mini gap remains roughly 100–250 Elo. Both models exceed 2500 Elo at depth 5, and unlike the unanchored case, Base is consistently stronger than Mini across all settings. Anchoring also gives Base a steady 10–30 Elo improvement, supporting the claim that shifting relative to the Black mean provides a more reliable evaluation coordinate.

4.2 EQUAL-DEPTH SIMULATIONS

We next evaluate SOLIS in direct equal-depth matches against Stockfish. Guided by the Elo scaling results in Section 4.1, we fix the SOLIS branching factor at $W=3$ because wider searches yield only marginal gains, demonstrating that the learned scorer already concentrates strong continuations

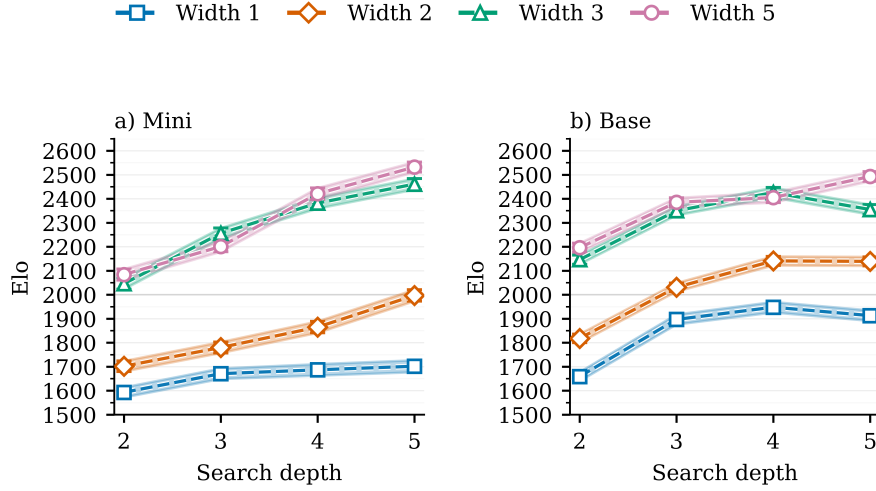


Figure 3: Elo estimates for SOLIS Mini and Base across search widths and depths using *unanchored* planning. Shaded bands denote 95% confidence intervals.

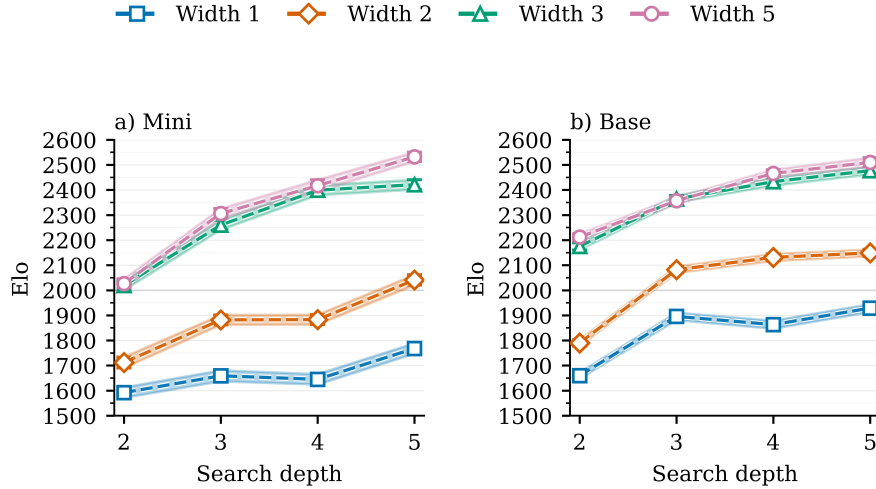


Figure 4: Elo estimates for SOLIS Mini and Base across search widths and depths using *anchored* planning. Shaded bands denote 95% confidence intervals.

near the top. Stockfish uses its default expansion without a fixed branching factor. For each depth $S \in \{3, 4, 5\}$, we simulate 200 games with alternating colors. Table 2 reports the resulting match points for both unanchored and anchored scoring.

Simulation results With unanchored scoring, SOLIS Base scores more match points than Stockfish at every tested depth, while Mini is competitive at depths 3 and 5 and trails at depth 4. With anchored scoring, Base loses narrowly at depth 3 but wins at depths 4 and 5; Mini also wins at depths 3 and 5 while falling behind at depth 4. These head-to-head results confirm that embedding-guided search is not merely rating-competitive under caps, but also viable in equal-depth play against the Stockfish configuration used here.

Nodes searched For SOLIS with a fixed branching factor of $W=3$, the maximum nodes searched at depth S is $N(S) = \sum_{i=0}^S 3^i$, giving $N(3) = 40$, $N(4) = 121$, and $N(5) = 364$. Stockfish, by contrast, explores a variable number of nodes per move depending on pruning and heuristics:

Table 2: Equal-depth match points for SOLIS vs. Stockfish. The middle block reports direct-similarity planning with the global advantage vector, and the right block reports anchored projection. Higher is better; draws count as 0.5.

Config	Depth	Direct Similarity		Anchored Projection	
		SOLIS score	Stockfish score	SOLIS score	Stockfish score
Base	3	106	94	98	102
	4	109	91	108	92
	5	107.5	92.5	110	90
Mini	3	108	92	105	95
	4	85	115	91	109
	5	101	99	113	87

at depth 3 it averages 137 nodes (maximum 1625), at depth 4 it averages 214 nodes (maximum 4463), and at depth 5 it averages 338 nodes (maximum 10611). SOLIS remains competitive while searching a small, fixed tree, showing that **higher-quality representations can recover much of the strength normally obtained through broad expansion.**

4.3 INTERPRETABLE GAME TRAJECTORIES

Latent game trajectories make the learned planning geometry directly inspectable. Figure 5 shows three representative games: a White win, a balanced draw, and a Black win. Each sequence is plotted on the same 2D projection, with independently embedded positions connected by arrows to show game progression. Decisive games trace coherent paths toward the appropriate winning region, whereas balanced games remain near the center. These trajectories extend Figure 1 by showing that SOLIS’ evaluation-aligned geometry governs not only static positions but also the temporal flow of actual games.

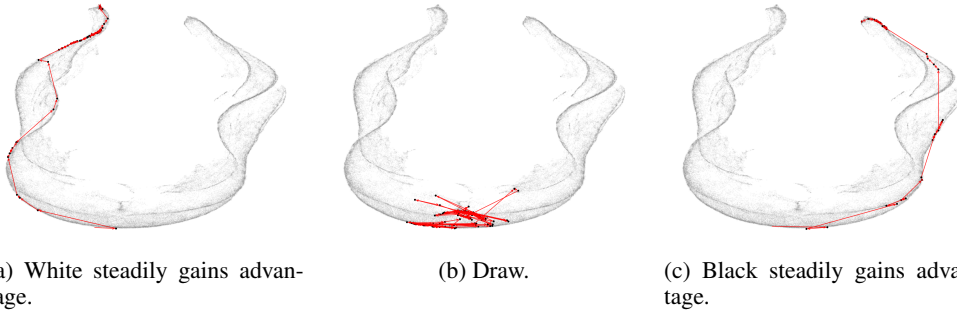


Figure 5: Latent trajectories of three games in the shared SOLIS representation space. Red arrows track position progression and reveal smooth movement toward winning regions in decisive games.

5 DISCUSSION

5.1 LIMITATIONS

Blindness to threefold repetition SOLIS’ strength comes from decisive alignment with an advantage vector, which keeps inference lightweight but deliberately omits move history. That omission creates a concrete failure mode: threefold repetition. If the same position, including side to move, castling rights, and en-passant rights, occurs three times during a game, either player can claim a draw. A prepared opponent can therefore force repeats, such as perpetual checks, to neutralize winning positions that SOLIS would otherwise be able to convert. Game C in Appendix A.4 illustrates this limitation: SOLIS reaches a winning position against Stockfish but permits an avoidable repetition draw.

Indecisiveness in winning positions SOLIS also treats winning positions with identical win probability as equivalent, even when their forcing lengths differ. A mate-in-1 and a mate-in-20 both receive win probability 1.0, although the shorter line is strategically preferable. When SOLIS selects a longer winning continuation, it gives the opponent additional opportunities to complicate the position, which can produce unnecessary draws or occasional losses. Ruoss et al. (2024) observed the same issue and avoided it by consulting Stockfish as an external oracle. SOLIS instead follows its own learned solution path, and adding preference for shorter forced wins is a clear next improvement. Game 1 in Appendix A.5 shows a case in which SOLIS has many winning continuations for several moves but chooses suboptimal moves that lead to a draw.

5.2 CONCLUSION

We demonstrate that planning can be performed directly in an evaluation-aligned latent space without training a policy or value head. SOLIS learns a contrastive embedding in which proximity reflects outcome similarity and a single global *advantage vector* points from losing to winning regions. Planning is therefore reduced to vector arithmetic: rank candidate actions by alignment with this direction and expand only a small search tree.

In chess, SOLIS operates with a fixed branching factor $W=3$ and shallow depths $S \in \{3, 4, 5\}$. Across these settings, it matches or exceeds the time-capped Stockfish configuration in several equal-depth matchups while searching far fewer nodes per move (40–364 for SOLIS versus hundreds to thousands for Stockfish at the same depths). Elo increases with depth; with anchored scoring, both Mini and Base exceed 2500 Elo at depth five, and anchoring gives Base a consistent 10–30 Elo gain. Additional width contributes little beyond three candidates, demonstrating that the learned space supplies the core search efficiency. The same embedding also yields readable game paths: decisive games move steadily toward winning regions, while draws remain near neutral.

These results make evaluation-aligned latent planning a compelling and compute-efficient alternative to rollout-heavy search or explicit dynamics modeling. The principle should transfer to other perfect-information games and to control tasks with large action spaces where low-branching search is valuable. The next steps are clear: add rule-aware state features, prefer shorter forced wins, combine SOLIS with policy priors or MCTS, study calibration and uncertainty, and test transfer beyond chess.

REPRODUCIBILITY STATEMENT

Appendix A.2 provides the complete training setup, including hyperparameters and implementation details. Section 3 specifies the encoder architecture, input representation, supervised contrastive objective, and inference procedure, with model configurations summarized in Table 1. Section 4 and Appendix A.3 define the evaluation protocol, Stockfish configuration, and rating procedure. Appendix A.6 reports the detailed match outcomes needed to verify the Elo estimates.

REFERENCES

- T. Chen, S. Kornblith, M. Norouzi, and G. Hinton. A simple framework for contrastive learning of visual representations, 2020. URL <https://arxiv.org/abs/2002.05709>.
- S. Chowdhury, G. Hamerly, and M. McGarrity. Active learning strategy using contrastive learning and k-means for aquatic invasive species recognition. In *2024 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, pages 848–858, 2024. doi: 10.1109/WACVW60836.2024.00097.
- R. Coulom. Whole-history rating: A bayesian rating system for players of time-varying strength. In *Computers and Games*, volume 5131 of *LNC3*, pages 113–124. Springer, 2008. doi: 10.1007/978-3-540-87608-3_10.
- J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. Bert: Pre-training of deep bidirectional transformers for language understanding, 2019. URL <https://arxiv.org/abs/1810.04805>.

486 A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani,
487 M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby. An image is worth 16x16 words:
488 Transformers for image recognition at scale, 2021. URL [https://arxiv.org/abs/2010.](https://arxiv.org/abs/2010.11929)
489 11929.

490 R. Gieselmann and F. T. Pokorny. Latent planning via expansive tree search. In A. H. Oh, A. Agar-
491 wal, D. Belgrave, and K. Cho, editors, *Advances in Neural Information Processing Systems*, 2022.
492 URL <https://openreview.net/forum?id=zSdz5scsnzU>.

493 R. Girdhar, A. El-Nouby, Z. Liu, M. Singh, K. V. Alwala, A. Joulin, and I. Misra. Imagebind: One
494 embedding space to bind them all, 2023. URL <https://arxiv.org/abs/2305.05665>.

495 D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, and J. Davidson. Learning latent
496 dynamics for planning from pixels, 2019. URL <https://arxiv.org/abs/1811.04551>.

497 D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi. Dream to control: Learning behaviors by latent
498 imagination, 2020. URL <https://arxiv.org/abs/1912.01603>.

499 D. Hafner, T. Lillicrap, M. Norouzi, and J. Ba. Mastering atari with discrete world models, 2022.
500 URL <https://arxiv.org/abs/2010.02193>.

501 D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering diverse domains through world models,
502 2024. URL <https://arxiv.org/abs/2301.04104>.

503 J. B. Hamrick, V. Bapst, A. Sanchez-Gonzalez, T. Pfaff, T. Weber, L. Buesing, and P. W.
504 Battaglia. Combining q-learning and search with amortized value estimates, 2020. URL
505 <https://arxiv.org/abs/1912.02807>.

506 K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition, 2015. URL
507 <https://arxiv.org/abs/1512.03385>.

508 D. Hendrycks and K. Gimpel. Gaussian error linear units (gelus), 2023. URL [https://arxiv.](https://arxiv.org/abs/1606.08415)
509 [org/abs/1606.08415](https://arxiv.org/abs/1606.08415).

510 M. Janner, J. Fu, M. Zhang, and S. Levine. When to trust your model: Model-based policy opti-
511 mization, 2021. URL <https://arxiv.org/abs/1906.08253>.

512 Z. Jiang, T. Zhang, M. Janner, Y. Li, T. Rocktäschel, E. Grefenstette, and Y. Tian. Efficient planning
513 in a compact latent action space, 2023. URL <https://arxiv.org/abs/2208.10291>.

514 P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan.
515 Supervised contrastive learning, 2021. URL <https://arxiv.org/abs/2004.11362>.

516 M. Lai. Giraffe: Using deep reinforcement learning to play chess, 2015. URL [https://arxiv.](https://arxiv.org/abs/1509.01549)
517 [org/abs/1509.01549](https://arxiv.org/abs/1509.01549).

518 L. McInnes, J. Healy, and J. Melville. Umap: Uniform manifold approximation and projection for
519 dimension reduction, 2020. URL <https://arxiv.org/abs/1802.03426>.

520 T. M. Moerland, J. Broekens, A. Plaat, and C. M. Jonker. Model-based reinforcement learning: A
521 survey, 2022. URL <https://arxiv.org/abs/2006.16712>.

522 Y. Nasu. Efficiently updatable neural-network-based evaluation functions for computer shogi, 2018.
523 Accessed: 2024-04-25.

524 G.-C. Pascutto and G. Linscott. Leelachesszero, 2018. URL <https://lczero.org>. Accessed:
525 2024-04-25.

526 A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein,
527 L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy,
528 B. Steiner, L. Fang, J. Bai, and S. Chintala. Pytorch: An imperative style, high-performance deep
529 learning library, 2019. URL <https://arxiv.org/abs/1912.01703>.

-
- M. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. Wiley Series in Probability and Statistics. Wiley, 2014. ISBN 9781118625873. URL <https://books.google.com/books?id=VvBjBAAQBAJ>.
- A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever. Learning transferable visual models from natural language supervision, 2021. URL <https://arxiv.org/abs/2103.00020>.
- T. Romstad, M. Costalba, J. Kiiski, G. Linscott, Y. Nasu, M. Isozaki, and H. Noda. Stockfish, 2008. URL <https://stockfishchess.org>. Accessed: 2025-04-22.
- A. Ruoss, G. Delétang, S. Medapati, J. Grau-Moya, L. K. Wenliang, E. Catt, J. Reid, C. A. Lewis, J. Veness, and T. Genewein. Amortized planning with large-scale transformers: A case study on chess, 2024. URL <https://arxiv.org/abs/2402.04494>.
- S. Russell and P. Norvig. *Artificial Intelligence, A modern approach*. 1995.
- J. Schrittwieser, I. Antonoglou, T. Hubert, K. Simonyan, L. Sifre, S. Schmitt, A. Guez, E. Lockhart, D. Hassabis, T. Graepel, T. Lillicrap, and D. Silver. Mastering atari, go, chess and shogi by planning with a learned model. *Nature*, 588(7839):604–609, Dec. 2020. ISSN 1476-4687. doi: 10.1038/s41586-020-03051-4. URL <http://dx.doi.org/10.1038/s41586-020-03051-4>.
- C. E. Shannon. XXII. programming a computer for playing chess. *The London, Edinburgh, and Dublin Philosophical Magazine and Journal of Science*, 41(314):256–275, 1950.
- D. Silver, T. Hubert, J. Schrittwieser, I. Antonoglou, M. Lai, A. Guez, M. Lanctot, L. Sifre, D. Kumar, T. Graepel, T. Lillicrap, K. Simonyan, and D. Hassabis. Mastering chess and shogi by self-play with a general reinforcement learning algorithm, 2017. URL <https://arxiv.org/abs/1712.01815>.
- N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.*, 15(1):1929–1958, Jan. 2014. ISSN 1532-4435.
- R. S. Sutton. Dyna, an integrated architecture for learning, planning, and reacting. *ACM Sigart Bulletin*, 2(4):160–163, 1991.
- R. S. Sutton and A. G. Barto. *Reinforcement learning: An introduction*, volume 1. MIT press Cambridge, 1998.
- J. v. Neumann. Zur theorie der gesellschaftsspiele. *Mathematische annalen*, 100(1):295–320, 1928.
- A. van den Oord, Y. Li, and O. Vinyals. Representation learning with contrastive predictive coding, 2019. URL <https://arxiv.org/abs/1807.03748>.
- A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need, 2023. URL <https://arxiv.org/abs/1706.03762>.
- A. L. Zobrist. A new hashing method with application for game playing. *ICGA Journal*, 13(2):69–73, 1990.

A APPENDIX

A.1 USE OF LARGE LANGUAGE MODELS (LLMs)

Large language models were used only for minor support tasks, including formatting and figure/table placement. They were not used to design experiments, analyze results, or generate technical content. All methodological and experimental contributions were developed and verified by the authors.

A.2 TRAINING SETUP

Both SOLIS configurations, Mini and Base, are trained with supervised contrastive learning on 5M ChessBench positions. Training uses cosine similarity over ℓ_2 -normalized embeddings and SGD with momentum on 6 48 GB L40S GPUs for 400k steps with batch size 128.

The two most important settings are τ and δ . The temperature $\tau = 0.07$ follows prior contrastive work (Radford et al., 2021; Girdhar et al., 2023): smaller values make similarity scores sharper, whereas larger values oversmooth gradients. Across $\tau \in \{0.05, 0.07, 0.1\}$, 0.07 converged most reliably for both model sizes. The margin $\delta = 0.05$ sets the evaluation tolerance for positives, balancing excessive neighborhood collapse at large δ against excessive fragmentation at small δ . It also matches the practical granularity of Stockfish evaluations, where sub-0.05 differences are rarely meaningful.

Table 3: Training hyperparameters used for both SOLIS configurations.

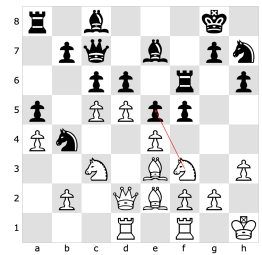
Attribute	Value
Loss	SupCon
τ	0.07
δ (positive margin)	0.05
Positives per anchor	5
Optimizer	SGD
Momentum	0.9
Learning rate	0.05
Batch size	128
Steps	400k
Dropout	0.10
Activation	GELU
Similarity	cosine
Embedding norm	ℓ_2
Token length	77

A.3 STOCKFISH SETUP

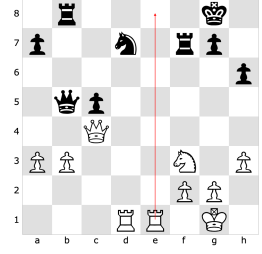
All experiments use the official Linux release of Stockfish 16 (Romstad et al., 2008) with a 32 MB hash and default threading. Rating caps are set with `UCI_LimitStrength` and `UCI_Elo`; fixed-depth runs use the `depth` parameter. To measure nodes searched for each move, we set the engine information level to `INFO_ALL` and extract the `nodes` attribute.

A.4 REPRESENTATIVE GAMES: STOCKFISH (CAP) VS. SOLIS (DEPTH, WIDTH)

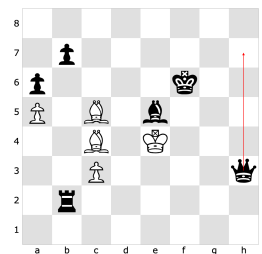
We include representative games between SOLIS and Stockfish in which SOLIS is limited to search depth S and width W , while Stockfish uses unbounded search under a rating cap. Games *A*, *B*, and *E* demonstrate SOLIS' tactical execution and long-term compensation. Games *C* and *D* expose the specific limitations discussed in the main text. In each table, the left column gives the game in Portable Game Notation (PGN), and the right column shows a critical position.

<pre> 1. d4 Nf6 2. c4 d6 3. Nc3 e5 4. Nf3 Qe7 5. e4 c6 6. d5 Qc7 7. Be2 h6 8. a4 a5 9. O-O Be7 10. h3 Na6 11. Be3 Nb4 12. Kh1 Nh7 13. Qd2 O-O 14. Rad1 f5 15. c5 Rf6 16. Nxe5 f4 17. Bxf4 Rxf4 18. Qxf4 dxe5 19. d6 exf4 20. Bc4+ Kf8 21. dxc7 Ke8 22. e5 Ng5 23. Rd6 Bxd6 24. exd6 Bd7 25. Re1+ Kf8 26. Re7 Bf5 27. h4 Nd5 28. Nxd5 cxd5 29. Bxd5 Be6 30. Bxe6 f3 31. hxg5 b6 32. Bh3 hxg5 33. Re5 Rc8 34. Rf5+ Kg8 35. Rf7 g4 36. cxb6 g6 37. d7 fxg2+ 38. Kh2 Rf8 39. Bxg2 Kxf7 40. Bc6 Ke7 41. Kg3 Rf3+ 42. Kh2 Rd3 43. b7 Rh3+ 44. Kg2 Re3 45. d8=Q+ Kf7 46. b8=Q Rg3+ 47. Kh2 Rh3+ 48. Kg1 Rh7 49. Qe8+ Kf6 50. Qbd8+ Kf5 51. Be4+ Kf4 52. Qf6# 1-0 </pre>	
--	---

Game A — SOLIS ($S=5$, $W=3$) vs. SF 2350, SOLIS–White. Result: 1–0. SOLIS makes a purposeful knight sacrifice and converts the resulting middlegame initiative.

<pre> 1. d4 d5 2. e3 c6 3. Bd3 h6 4. h3 Nf6 5. Nf3 e6 6. O-O b6 7. c4 Ba6 8. b3 Nbd7 9. Bb2 Bb4 10. a3 Be7 11. Nbd2 O-O 12. Re1 c5 13. Qe2 Qc7 14. e4 Nxe4 15. Nxe4 dxe4 16. Bxe4 Rab8 17. d5 Bf6 18. dxe6 Bxb2 19. exf7+ Rxf7 20. Qxb2 Bb7 21. Bxb7 Qxb7 22. Rad1 b5 23. Qe2 bxc4 24. Qxc4 Qb5 25. Re8+ Nf8 26. Rxb8 Qxb8 27. Rd5 Qf4 28. Qxf4 Rxf4 29. Rxc5 Ne6 30. Rc6 Nd4 31. Nxd4 a5 32. Rc4 Rf8 33. Nc6 Kh8 34. Nxa5 Ra8 35. b4 Rd8 36. g3 Rd3 37. a4 g5 38. b5 Rd2 39. b6 Rd5 40. b7 Rxa5 41. Rc8+ Kh7 42. b8=Q Rd5 43. Rc7+ Kg6 44. Qg8+ Kf5 45. Qxd5+ Kg6 46. Qf7# 1-0 </pre>	
--	--

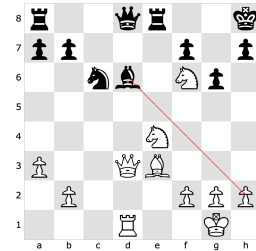
Game B — SOLIS ($S=5$, $W=3$) vs. SF 2300, SOLIS–White. Result: 1–0. SOLIS sacrifices a rook to win Stockfish's queen and decisively converts the advantage.

<pre> 1. e3 Nf6 2. Nc3 d5 3. Nf3 c5 4. d4 e6 5. g3 a6 6. a4 Nc6 7. Ne2 Bd6 8. Bg2 O-O 9. h3 Re8 10. c3 Qc7 11. Nd2 e5 12. Nb3 c4 13. dxe5 Nxe5 14. Nbd4 Nd3+ 15. Kf1 Nc5 16. Bf3 Bd7 17. Kg2 Be5 18. a5 Rad8 19. Qc2 g6 20. Bd2 h5 21. Be1 h4 22. gxh4 Kg7 23. Ng3 Rh8 24. h5 Rdg8 25. hxg6 fxg6 26. Be2 Qd6 27. Qd1 Qe7 28. f3 Bd6 29. Nf1 Rh5 30. h4 Rgh8 31. Bf2 R5h7 32. Qe1 Bb8 33. Bg3 Ba7 34. Bf2 Nh5 35. Ng3 Bb8 36. f4 Nf6 37. h5 Rh6 38. hxg6 Rxh1 39. Qxh1 Qe8 40. Ndf5+ Bxf5 41. Nxf5+ Kxg6 42. Ne7+ Kf7 43. e4 Rxh1 44. Rxh1 Qxe7 45. Bxc5 Qxe4+ 46. Bf3 Qg6+ 47. Kf1 Bxf4 48. b4 cxb3 49. Rh8 Ke6 50. Rf8 b2 51. Rxf6+ Kxf6 52. Bxd5 b1=R+ 53. Ke2 Rb2+ 54. Kf3 Be5 55. Bc4 Qh5+ 56. Ke3 Qh3+ 57. Ke4 Qh7+ 58. Kf3 Qh3+ 59. Ke4 Qh7+ 60. Kf3 Qh3+ 61. Ke4 Qh7+ 62. Kf3 Qh3+ 63. Ke4 Qh7+ 64. Kf3 Qh3+ 65. Ke4 1/2-1/2 </pre>	
--	---

Game C — SF 2200 vs. SOLIS ($S=5$, $W=3$), SOLIS–Black. Result: 1/2–1/2. SOLIS reaches a winning endgame but permits a draw because the representation omits threefold-repetition history.

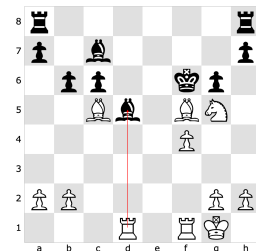
702
703
704
705
706
707
708
709
710
711
712
713
714
715
716
717
718
719
720
721
722
723
724
725
726
727
728
729
730
731
732
733
734
735
736
737
738
739
740
741
742
743
744
745
746
747
748
749
750
751
752
753
754
755

1. d4 d5 2. c4 dxc4 3. e3 e5 4. Bxc4 exd4 5. exd4
Nf6 6. Nc3 Bd6 7. Qe2+ Qe7 8. Be3 O-O 9. Nf3 c6 10.
O-O Be6 11. Rfd1 Re8 12. d5 cxd5 13. Bxd5 Nc6 14.
a3 Nxd5 15. Nxd5 Qd8 16. Qd3 Bg4 17. Ng5 g6 18. Ne4
Bxd1 19. Ndf6+ Kh8 20. Rxd1 Bxh2+ 21. Kxh2 Qxd3 22.
Rxd3 Red8 23. Nd7 Kg7 24. f3 h6 25. Rd6 f5 26. Nec5
Re8 27. Bd2 Re2 28. Kh1 Kf7 29. b3 Ke7 30. Nxb7 Rc8
31. Ndc5 Kf7 32. Kg1 Kg8 33. Rxg6+ Kf7 34. Rd6 Ne7
35. Rd7 Rg8 36. Nd6+ Kg6 37. Nc4 Kh5 38. Ne3 f4 39.
Ne6 fxe3 40. Nf4+ Kg5 41. Nxe2 exd2 42. Kf2 Kf6 43.
Nc3 Ke6 44. Rd4 Rc8 45. Ne4 d1=N+ 46. Rxd1 Rc2+ 47.
Kg1 Nf5 48. Rd2 Rxd2 49. Nxd2 Ke5 50. Kh2 Kf4 51.
b4 Ke3 52. Ne4 Nd4 53. a4 Nc2 54. b5 Nd4 55. Nd6
Ne6 56. a5 Kd3 57. g4 Kd4 58. b6 axb6 59. axb6 Nc5
60. Kg3 Kd5 61. b7 Na6 62. Nf7 Kc6 63. Nxh6 Kxb7
64. f4 Kc7 65. Kh4 Kd7 66. Kh5 Nc5 67. f5 Kd6 68.
f6 Ne6 69. Nf7+ Kc6 70. Kh6 Kd5 71. g5 Nc5 72. Kg6
Ne6 73. Kh5 Kd4 74. g6 Kc4 75. Ne5+ Kd5 76. Ng4 Ke4
77. Kh6 Kf5 78. g7 Nxg7 79. Kxg7 Kxg4 80. Kf7 Kf5
81. Ke7 Kg5 82. f7 Kf4 83. Kf6 Kf3 84. Ke5 Kg3 85.
Ke4 Kf2 86. Kd3 Kg1 87. Ke3 Kh2 88. f8=Q Kg3 89.
Qg8+ Kh3 90. Kf4 Kh2 91. Kf3 Kh3 92. Qg3# 1-0



Game D — SF 2500 vs. SOLIS ($S=5, W=3$), SOLIS–Black. Result: 1-0 SOLIS chooses an insufficiently grounded piece sacrifice and falls into a long-term material disadvantage.

1. d4 d5 2. e3 Nd7 3. c4 e5 4. dxe5 dxc4 5. Nf3
Qe7 6. Bxc4 Nxe5 7. Nxe5 Qxe5 8. Qb3 Qh5 9. e4
Nf6 10. O-O Bd6 11. Qb5+ c6 12. Qxh5 Nxh5 13. Nc3
Be5 14. f4 Bc7 15. Be3 g6 16. Bb3 Be6 17. Rad1 Ng7
18. Bc2 f5 19. Bc5 Kf7 20. exf5 Nxf5 21. Ne4 b6
22. Ng5+ Kf6 23. Bxf5 Bd5 24. Rxd5 cxd5 25. Bd4+
Kxf5 26. h3 Bxf4 27. Bxh8 Kxg5 28. Bg7 Rg8 29. Bd4
Re8 30. Rf3 Re1+ 31. Kf2 Re6 32. g3 Bd2 33. Rf7
Re4 34. Bf6+ Kh6 35. g4 a6 36. Kg3 d4 37. h4 Be1+
38. Kh3 Re3+ 39. Kg2 Bxh4 40. Bxh4 a5 41. Bf6 a4
42. Rd7 Rd3 43. Rd5 b5 44. Rxb5 a3 45. Re5 Rd2+
46. Kf3 axb2 47. g5+ Kh5 48. Rb5 Rd3+ 49. Kf4 h6
50. gxh6+ Kxh6 51. Rxb2 g5+ 52. Kg4 Rd1 53. Bxg5+
Kg6 54. Rb6+ Kg7 55. Kf5 Kf7 56. Rb7+ Ke8 57. a4
Ra1 58. Re7+ Kf8 59. Rh7 Rxa4 60. Bh6+ Kg8 61. Re7
Rc4 62. Kg5 Rc5+ 63. Kg4 Ra5 64. Rg7+ Kh8 65. Re7
d3 66. Rd7 d2 67. Bxd2 Rc5 68. Bf4 Kg8 69. Be3 Rb5
70. Bh6 Re5 71. Kf4 Rc5 72. Ke4 Rb5 73. Kf4 Rb4+
74. Ke5 Rb1 75. Kf6 Rb2 76. Kg6 Rb6+ 77. Kh5 Rb2
78. Bg5 Rb6 79. Bh6 Re6 80. Kg5 Ra6 81. Rg7+ Kh8
82. Re7 Ra5+ 83. Kg6 Ra6+ 84. Kh5 Kg8 85. Rb7 Rc6
86. Rd7 Rc5+ 87. Bg5 Rc2 88. Kg6 Rc6+ 89. Bf6 Re6
90. Rd8+ Re8 91. Rxe8# 1-0

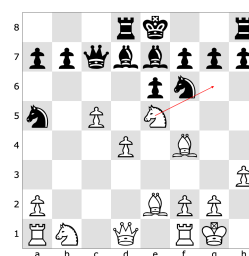


Game E — SOLIS ($S=5, W=3$) vs. Stockfish 2500 , SOLIS–White. Result: 1-0 SOLIS executes an exchange sacrifice for long-term compensation and converts the advantage into a win.

A.5 EQUAL-DEPTH MATCHES: SOLIS D vs. STOCKFISH D

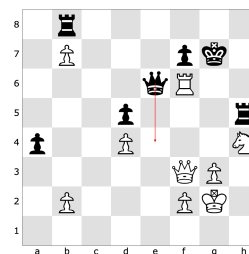
We include illustrative equal-depth games in which SOLIS and Stockfish both search to depth D . These examples show SOLIS' latent planner competing directly without a Stockfish rating cap. In each table, the left column records the game in Portable Game Notation (PGN), and the right column highlights a critical position.

1. d4 d5 2. e3 c5 3. Nf3 Nf6 4. Be2 e6 5. O-O Nc6 6. b3 cxd4 7. exd4 Be7 8. c3 Qc7 9. h3 Bd7 10. c4 dxc4 11. bxc4 Na5 12. Ne5 Rd8 13. Bf4 Bd6 14. c5 Be7 15. Ng6 Qc8 16. Nxh8 Bc6 17. Nc3 Kf8 18. Rc1 Qa8 19. Bc7 b5 20. Bxa5 Bxg2 21. d5 Bxh3 22. Qd3 Bf5 23. Qg3 b4 24. Rfd1 Bxc5 25. Qc7 Be7 26. Nb5 Nxd5 27. Rxd5 Qxd5 28. Bxb4 Qd7 29. Qxd7 Re8 30. Rc8 g5 31. Qxe7+ Kg7 32. Qxf7+ Kh6 33. Bf8+ Rxf8 34. Qxf8# 1-0



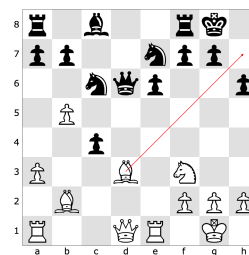
Game F — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 SOLIS sacrifices a knight in the middlegame to force a concrete tactical advantage.

1. d4 d5 2. e3 c5 3. Nf3 e6 4. Be2 cxd4 5. exd4 Bd6 6. O-O Nf6 7. Bg5 Nbd7 8. Nbd2 h6 9. Bh4 O-O 10. Re1 a5 11. Bf1 Bf4 12. c3 Qb6 13. Qc2 Qc7 14. a4 b6 15. Bb5 Ba6 16. Bxa6 Rxa6 17. Qd3 Ra7 18. Re2 Qc8 19. Reel Nh5 20. Nf1 Nh6 21. N1d2 Nh5 22. Nf1 Bc7 23. Qd1 Nh6 24. Qd3 Nh5 25. Qd1 Nh6 26. Qd3 Rb7 27. Qb5 Ra7 28. N1d2 Nh5 29. h3 g5 30. Rac1 gxh4 31. c4 Nf4 32. Rc3 Qd8 33. g3 Nxh3+ 34. Kg2 Ng5 35. cxd5 h3+ 36. Kg1 h2+ 37. Kxh2 exd5 38. Rc6 Nf6 39. Nxg5 hxg5 40. Kg2 Re8 41. Rh1 Bd6 42. Nf3 Kg7 43. Nxg5 Rc7 44. Rxd6 Qxd6 45. Qf1 b5 46. axb5 Rcc8 47. Rh4 Rh8 48. Nf3 Ne4 49. Rf4 Rh5 50. Qe2 Qe6 51. Nh4 Rch8 52. b6 Nf6 53. Qf3 a4 54. b7 Rb8 55. Rxf6 Qe4 56. Rxf7+ Kh8 57. Qxe4 dxe4 58. Rc7 Rg5 59. Rc8+ Rg8 60. Rc7 Rg5 61. Rc8+ Rg8 62. Ng6+ Kg7 63. Ne7 Rf8 64. Rc7 Rf7 65. d5 Rbf8 66. Nf5+ Kg6 67. Rxf7 Kxf7 68. Nd6+ Kg6 69. Nc8 Kg5 70. b8=Q Kg4 71. Qc7 Rf5 72. Nd6 e3 73. Qg7+ Rg5 74. f3+ Kh5 75. Qh7# 1-0



Game G — SOLIS $S = 5$ vs. Stockfish $S = 5$, SOLIS-White. Result: 1-0 Stockfish builds an advantage but blunders in the late middlegame; SOLIS immediately capitalizes and wins the endgame.

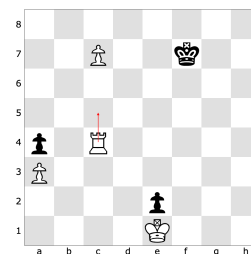
1. d4 c6 2. e4 d5 3. Nc3 Nf6 4. e5 Ng8 5. Bd3 e6 6. Nf3 c5 7. dxc5 Bxc5 8. O-O Nc6 9. a3 h6 10. b4 Bb6 11. Nb5 Bc7 12. Re1 Bb8 13. c4 Nge7 14. Nd6+ Bxd6 15. exd6 Qxd6 16. Bb2 O-O 17. b5 dxc4 18. Bh7+ Kh8 19. Qxd6 f6 20. Be4 c3 21. Bxc3 e5 22. bxc6 Bf5 23. Qxe7 Bxe4 24. Rxe4 bxc6 25. Nxe5 Rfe8 26. Nf7+ Kg8 27. Nxh6+ Kh7 28. Rae1 Rad8 29. Ba5 Rd4 30. g4 Rxe4 31. Rxe4 Rb8 32. Qxa7 Rd8 33. Rf4 Rd3 34. g5 fxg5 35. Rd4 Rxa3 36. Kf1 Rb3 37. Rb4 Rc3 38. h4 gxh4 39. Rd4 h3 40. Qe7 h2 41. Qe4+ Kxh6 42. Rd6+ Kg5 43. Rg6+ Kh5 44. Qg4# 1-0



Game H — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1-0 SOLIS sacrifices a bishop to trigger a discovered attack on Stockfish's queen and converts cleanly.

810
811
812
813
814
815
816
817
818
819
820
821
822
823
824
825
826
827
828
829
830
831
832
833
834
835
836
837
838
839
840
841
842
843
844
845
846
847
848
849
850
851
852
853
854
855
856
857
858
859
860
861
862
863

1. d4 d5 2. e3 c6 3. c4 e6 4. Nf3 Nf6 5. Be2 c5
6. O-O Nc6 7. b3 a6 8. Bb2 Be7 9. a3 cxd4 10. exd4
O-O 11. c5 b6 12. b4 bxc5 13. dxc5 Bb7 14. Nbd2 a5
15. b5 Bxc5 16. Rc1 Qb6 17. bxc6 Ng4 18. Rxc5 Ba6
19. Rc2 Nxf2 20. Rxf2 Bxe2 21. Qxe2 a4 22. c7 Rac8
23. Bd4 Qb5 24. Qxb5 Rfe8 25. Qb8 h6 26. Ne5 g5 27.
h4 Rf8 28. Ne4 dxe4 29. Rxf7 Rxb8 30. Rxf8+ Rxf8 31.
hxcg5 Rc8 32. gxh6 e3 33. Bxe3 Kh7 34. Rc4 Kg8 35.
Kh2 Kh7 36. g4 Rg8 37. g5 Ra8 38. Rg4 Kh8 39. Ba7
Rf8 40. Bb8 Rf2+ 41. Kg1 Rc2 42. Rc4 Rb2 43. Ba7
Rb1+ 44. Kg2 Kh7 45. Bf2 Rb2 46. g6+ Kxh6 47. Kh2
Rxf2+ 48. Kg3 Rg2+ 49. Kxg2 Kg7 50. Rc1 Kf6 51. Kh2
Kg7 52. Nd7 e5 53. Nf6 Kxf6 54. Rg1 Kg7 55. Rf1 Kh6
56. Rg1 Kg7 57. Rf1 Kxg6 58. Rc1 Kg7 59. Rc4 Kf7
60. Kg2 Ke7 61. Kf2 Kd6 62. Ke1 Kd5 63. Rc1 Ke4 64.
Ke2 Kd4 65. Rc2 e4 66. Rc1 Kd5 67. Ke1 Ke5 68. Rc4
Kd5 69. Rc2 Kd6 70. Rc1 Ke7 71. Rc4 Kf7 72. Rc5 e3
73. Kf1 Ke7 74. Rc4 Kf7 75. Kg2 e2 76. Kf2 Ke7 77.
Ke1 Kf7 78. Rc5 Kg7 79. Rc2 Kf6 80. Rc4 Kf7 81. Rc5
Kg6 82. Kd2 Kf7 83. Ke1 Ke7 84. Rc4 Kf7 85. Rc5 Kg6
86. Rc4 Kf7 87. Rc5 1/2-1/2



Game I — SOLIS $S = 3$ vs. Stockfish $S = 3$, SOLIS-White. Result: 1/2-1/2 SOLIS exposes its forced-win-length limitation in a position where many continuations remain winning.

A.6 SOLIS VS. STOCKFISH RESULTS

For reproducibility, we report the complete SOLIS–Stockfish match outcomes used for Elo estimation, with SOLIS searching at depths $S \in \{2, 3, 4, 5\}$. Each table is indexed by Stockfish Elo cap in the rows and SOLIS search width W in the columns. Entries report win–loss–draw counts for each matchup, with total match points in parentheses. We provide separate tables for Base and Mini, and for both unanchored and anchored scoring.

Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–

Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 6: SOLIS Base vs. Stockfish with unanchored planning, reported by depth and width. Entries give W–D–L and total score out of 100.

918
919
920
921
922
923
924
925
926
927
928
929
930
931
932
933
934
935
936
937
938
939
940
941
942
943
944
945
946
947
948
949
950
951
952
953
954
955
956
957
958
959
960
961
962
963
964
965
966
967
968
969
970
971

Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	32–32–36 (48)	27–45–28 (50)	–	–	1350	–	–	–	–
1400	30–41–29 (50)	33–27–40 (46)	–	–	1400	–	–	–	–
1450	29–44–27 (51)	33–40–27 (53)	–	–	1450	–	–	–	–
1500	31–38–31 (50)	31–39–30 (50)	–	–	1500	29–38–33 (48)	28–43–29 (50)	–	–
1550	38–35–27 (56)	23–50–27 (48)	–	–	1550	41–36–23 (59)	36–37–27 (54)	–	–
1600	34–30–36 (49)	29–37–34 (48)	–	–	1600	37–30–33 (52)	37–30–33 (52)	–	–
1650	24–45–31 (46)	28–36–36 (46)	–	–	1650	32–38–30 (51)	30–40–30 (50)	–	–
1700	–	–	–	–	1700	28–37–35 (46)	21–43–36 (42)	–	–
1750	–	–	–	–	1750	36–33–31 (52)	28–40–32 (48)	–	–
1800	–	–	37–31–32 (52)	37–36–27 (55)	1800	39–24–37 (51)	29–35–36 (46)	–	–
1850	–	–	34–24–42 (46)	34–31–35 (50)	1850	–	–	–	–
1900	–	–	40–28–32 (54)	40–29–31 (54)	1900	–	–	–	–
1950	–	–	39–29–32 (54)	39–29–32 (54)	1950	–	–	–	–
2000	–	–	38–34–28 (55)	42–28–30 (56)	2000	–	–	34–35–31 (52)	38–21–41 (48)
2050	–	–	41–27–32 (54)	40–26–34 (53)	2050	–	–	38–34–28 (55)	34–33–33 (50)
2100	–	–	35–30–35 (50)	36–26–38 (49)	2100	–	–	33–34–33 (50)	34–29–37 (48)
2150	–	–	–	–	2150	–	–	31–35–34 (48)	38–25–37 (50)
2200	–	–	–	–	2200	–	–	38–18–44 (47)	31–26–43 (44)
2250	–	–	–	–	2250	–	–	36–23–41 (48)	45–26–29 (58)
2300	–	–	–	–	2300	–	–	40–22–38 (51)	34–26–40 (47)
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2500	–	–	–	–	2500	–	–	–	–
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	–	–	–	–	1350	–	–	–	–
1400	–	–	–	–	1400	–	–	–	–
1450	–	–	–	–	1450	–	–	–	–
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	36–42–22 (57)	35–31–34 (50)	–	–	1600	–	–	–	–
1650	29–40–31 (49)	36–33–31 (52)	–	–	1650	–	–	–	–
1700	33–34–33 (50)	36–32–32 (52)	–	–	1700	39–30–31 (54)	34–34–32 (51)	–	–
1750	43–26–31 (56)	28–38–34 (47)	–	–	1750	37–29–34 (52)	32–30–38 (47)	–	–
1800	38–24–38 (50)	30–37–33 (48)	–	–	1800	42–20–38 (52)	33–31–36 (48)	–	–
1850	49–23–28 (60)	29–33–38 (46)	–	–	1850	36–24–40 (48)	43–28–29 (57)	–	–
1900	–	–	–	–	1900	45–26–29 (58)	46–31–23 (62)	–	–
1950	–	–	–	–	1950	40–22–38 (51)	37–35–28 (54)	–	–
2000	–	–	–	–	2000	–	–	–	–
2050	–	–	–	–	2050	–	–	–	–
2100	–	–	–	–	2100	–	–	–	–
2150	–	–	31–31–38 (46)	44–21–35 (54)	2150	–	–	–	–
2200	–	–	35–29–36 (50)	48–27–25 (62)	2200	–	–	27–37–36 (46)	33–38–29 (52)
2250	–	–	37–31–32 (52)	37–28–35 (51)	2250	–	–	30–33–37 (46)	38–31–31 (54)
2300	–	–	27–37–36 (46)	36–34–30 (53)	2300	–	–	38–37–25 (56)	33–39–28 (52)
2350	–	–	22–42–36 (43)	37–35–28 (54)	2350	–	–	32–29–39 (46)	29–36–35 (47)
2400	–	–	36–29–35 (50)	37–26–37 (50)	2400	–	–	30–26–44 (43)	30–32–38 (46)
2500	–	–	–	–	2500	–	–	26–34–40 (43)	43–25–32 (56)

Figure 7: SOLIS Mini vs. Stockfish with unanchored planning, reported by depth and width. Entries give W–D–L and total score out of 100.

972
973
974
975
976
977
978
979
980
981
982
983
984
985
986
987
988
989
990
991
992
993
994
995
996
997
998
999
1000
1001
1002
1003
1004
1005
1006
1007
1008
1009
1010
1011
1012
1013
1014
1015
1016
1017
1018
1019
1020
1021
1022
1023
1024
1025

Base: Depth 2					Base: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	32–46–22 (55)	41–47–12 (64)	–	–	1500	–	–	–	–
1550	39–38–23 (58)	37–50–13 (62)	–	–	1550	–	–	–	–
1600	32–44–24 (54)	33–51–16 (58)	–	–	1600	–	–	–	–
1650	33–41–26 (54)	29–46–25 (52)	–	–	1650	–	–	–	–
1700	34–38–28 (53)	22–53–25 (48)	–	–	1700	–	–	–	–
1750	37–35–28 (54)	30–49–21 (54)	–	–	1750	–	–	–	–
1800	29–42–29 (50)	36–42–22 (57)	–	–	1800	39–29–32 (54)	37–49–14 (62)	–	–
1850	–	–	–	–	1850	35–32–33 (51)	30–50–20 (55)	–	–
1900	–	–	–	–	1900	37–30–33 (52)	40–41–19 (60)	–	–
1950	–	–	–	–	1950	31–28–41 (45)	37–44–19 (59)	–	–
2000	–	–	34–40–26 (54)	43–31–26 (58)	2000	41–27–32 (54)	33–45–22 (56)	–	–
2050	–	–	46–38–16 (65)	36–38–26 (55)	2050	39–27–34 (52)	31–41–28 (52)	–	–
2100	–	–	34–45–21 (56)	34–31–35 (50)	2100	41–27–32 (54)	36–38–26 (55)	–	–
2150	–	–	46–31–23 (62)	30–43–27 (52)	2150	–	–	–	–
2200	–	–	31–35–34 (48)	35–28–37 (49)	2200	–	–	33–42–25 (54)	30–41–29 (50)
2250	–	–	34–34–32 (51)	40–29–31 (54)	2250	–	–	35–41–24 (56)	33–43–24 (54)
2300	–	–	32–32–36 (48)	37–34–29 (54)	2300	–	–	35–33–32 (52)	36–34–30 (53)
2350	–	–	–	–	2350	–	–	37–37–26 (56)	29–37–34 (48)
2400	–	–	–	–	2400	–	–	34–36–30 (52)	37–38–25 (56)
2450	–	–	–	–	2450	–	–	37–28–35 (51)	43–29–28 (58)
2500	–	–	–	–	2500	–	–	41–30–29 (56)	34–30–36 (49)
2550	–	–	–	–	2550	–	–	–	–
2600	–	–	–	–	2600	–	–	–	–
2650	–	–	–	–	2650	–	–	–	–
2700	–	–	–	–	2700	–	–	–	–

Base: Depth 4					Base: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1500	–	–	–	–	1500	–	–	–	–
1550	–	–	–	–	1550	–	–	–	–
1600	–	–	–	–	1600	–	–	–	–
1650	–	–	–	–	1650	–	–	–	–
1700	–	–	–	–	1700	–	–	–	–
1750	–	–	–	–	1750	–	–	–	–
1800	–	–	–	–	1800	–	–	–	–
1850	–	–	–	–	1850	–	–	–	–
1900	–	–	–	–	1900	–	–	–	–
1950	37–26–37 (50)	41–42–17 (62)	–	–	1950	36–34–30 (53)	38–41–21 (58)	–	–
2000	43–20–37 (53)	31–43–26 (52)	–	–	2000	31–28–41 (45)	27–52–21 (53)	–	–
2050	38–27–35 (52)	40–38–22 (59)	–	–	2050	41–24–35 (53)	36–37–27 (54)	–	–
2100	42–25–33 (54)	32–48–20 (56)	–	–	2100	43–23–34 (54)	36–39–25 (56)	–	–
2150	36–23–41 (48)	31–47–22 (54)	–	–	2150	42–23–35 (54)	27–45–28 (50)	–	–
2200	46–13–41 (52)	39–39–22 (58)	–	–	2200	42–21–37 (52)	23–51–26 (48)	–	–
2250	–	–	–	–	2250	–	–	–	–
2300	–	–	–	–	2300	–	–	–	–
2350	–	–	–	–	2350	–	–	–	–
2400	–	–	–	–	2400	–	–	–	–
2450	–	–	40–39–21 (60)	39–38–23 (58)	2450	–	–	25–46–29 (48)	43–30–27 (58)
2500	–	–	38–38–24 (57)	41–40–19 (61)	2500	–	–	40–36–24 (58)	34–44–22 (56)
2550	–	–	44–20–36 (54)	42–29–29 (56)	2550	–	–	40–31–29 (56)	38–38–24 (57)
2600	–	–	46–17–37 (54)	37–23–40 (48)	2600	–	–	46–25–29 (58)	33–37–30 (52)
2650	–	–	41–18–41 (50)	44–22–34 (55)	2650	–	–	47–17–36 (56)	30–35–35 (48)
2700	–	–	45–16–39 (53)	42–22–36 (53)	2700	–	–	38–25–37 (50)	45–21–34 (56)

Figure 8: SOLIS Base vs. Stockfish with anchored planning, reported by depth and width. Entries give W–D–L and total score out of 100.

1026
1027
1028
1029
1030
1031
1032
1033
1034
1035
1036
1037
1038
1039
1040
1041
1042
1043
1044
1045
1046
1047
1048
1049
1050
1051
1052
1053
1054
1055
1056
1057
1058
1059
1060
1061
1062
1063
1064
1065
1066
1067
1068
1069
1070
1071
1072
1073
1074
1075
1076
1077
1078
1079

Mini: Depth 2					Mini: Depth 3				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	36-32-32 (52)	39-30-31 (54)	-	-	1350	-	-	-	-
1400	23-49-28 (48)	30-32-38 (46)	-	-	1400	-	-	-	-
1450	33-37-30 (52)	24-47-29 (48)	-	-	1450	-	-	-	-
1500	34-38-28 (53)	25-40-35 (45)	-	-	1500	29-40-31 (49)	35-33-32 (52)	-	-
1550	28-38-34 (47)	40-39-21 (60)	-	-	1550	34-35-31 (52)	37-29-34 (52)	-	-
1600	23-47-30 (46)	28-42-30 (49)	-	-	1600	32-36-32 (50)	32-39-29 (52)	-	-
1650	35-29-36 (50)	38-37-25 (56)	-	-	1650	34-30-36 (49)	37-33-30 (54)	-	-
1700	-	-	-	-	1700	36-30-34 (51)	34-35-31 (52)	-	-
1750	-	-	-	-	1750	34-25-41 (46)	34-30-36 (49)	-	-
1800	-	-	35-37-28 (54)	40-28-32 (54)	1800	32-29-39 (46)	32-31-37 (48)	-	-
1850	-	-	32-32-36 (48)	44-29-27 (58)	1850	-	-	-	-
1900	-	-	37-30-33 (52)	45-21-34 (56)	1900	-	-	-	-
1950	-	-	43-30-27 (58)	36-29-35 (50)	1950	-	-	-	-
2000	-	-	35-31-34 (50)	43-27-30 (56)	2000	-	-	46-32-22 (62)	38-27-35 (52)
2050	-	-	37-29-34 (52)	45-27-28 (58)	2050	-	-	37-27-36 (50)	39-26-35 (52)
2100	-	-	36-27-37 (50)	42-34-24 (59)	2100	-	-	29-34-37 (46)	48-17-35 (56)
2150	-	-	-	-	2150	-	-	37-33-30 (54)	39-25-36 (52)
2200	-	-	-	-	2200	-	-	37-31-32 (52)	41-23-36 (52)
2250	-	-	-	-	2250	-	-	36-23-41 (48)	44-20-36 (54)
2300	-	-	-	-	2300	-	-	30-43-27 (52)	38-26-36 (51)
2350	-	-	-	-	2350	-	-	-	-
2400	-	-	-	-	2400	-	-	-	-
2500	-	-	-	-	2500	-	-	-	-
Mini: Depth 4					Mini: Depth 5				
SF Elo	W=1	W=2	W=3	W=5	SF Elo	W=1	W=2	W=3	W=5
1350	-	-	-	-	1350	-	-	-	-
1400	-	-	-	-	1400	-	-	-	-
1450	-	-	-	-	1450	-	-	-	-
1500	-	-	-	-	1500	-	-	-	-
1550	-	-	-	-	1550	-	-	-	-
1600	26-45-29 (48)	31-33-36 (48)	-	-	1600	-	-	-	-
1650	37-39-24 (56)	28-32-40 (44)	-	-	1650	-	-	-	-
1700	23-39-38 (42)	29-41-30 (50)	-	-	1700	37-36-27 (55)	34-29-37 (48)	-	-
1750	36-29-35 (50)	38-30-32 (53)	-	-	1750	42-23-35 (54)	37-33-30 (54)	-	-
1800	35-24-41 (47)	33-36-31 (51)	-	-	1800	36-28-36 (50)	31-35-34 (48)	-	-
1850	41-21-38 (52)	30-28-42 (44)	-	-	1850	38-21-41 (48)	33-32-35 (49)	-	-
1900	-	-	-	-	1900	37-27-36 (50)	37-37-26 (56)	-	-
1950	-	-	-	-	1950	43-18-39 (52)	38-24-38 (50)	-	-
2000	-	-	-	-	2000	-	-	-	-
2050	-	-	-	-	2050	-	-	-	-
2100	-	-	-	-	2100	-	-	-	-
2150	-	-	39-23-38 (50)	32-30-38 (47)	2150	-	-	-	-
2200	-	-	33-40-27 (53)	38-25-37 (50)	2200	-	-	30-40-30 (50)	32-34-34 (49)
2250	-	-	36-31-33 (52)	30-27-43 (44)	2250	-	-	29-43-28 (50)	49-26-25 (62)
2300	-	-	32-33-35 (48)	37-28-35 (51)	2300	-	-	31-30-39 (46)	30-35-35 (48)
2350	-	-	35-34-31 (52)	40-25-35 (52)	2350	-	-	26-32-42 (42)	40-34-26 (57)
2400	-	-	28-26-46 (41)	26-38-36 (45)	2400	-	-	39-32-29 (55)	31-32-37 (47)
2500	-	-	-	-	2500	-	-	30-29-41 (44)	28-41-31 (48)

Figure 9: SOLIS Mini vs. Stockfish with anchored planning, reported by depth and width. Entries give W-D-L and total score out of 100.